

Social Network Graph – Trace Tables

Given connections: A-B, A-C, B-D, B-E, C-F, E-F

1. BFS Trace Table

Step	Vertex Processed	Queue	Visited
1	A	B, C	A, B, C
2	B	C, D, E	A, B, C, D, E
3	C	D, E, F	A, B, C, D, E, F
4	D	E, F	A, B, C, D, E, F
5	E	F	A, B, C, D, E, F
6	F	Empty	A, B, C, D, E, F

BFS Result: A → B → C → D → E → F

2. DFS Trace Table

Step	Current Vertex	Action
1	A	Visit A, go to B
2	B	Visit B, go to D
3	D	Visit D, backtrack to B
4	B	Go to E
5	E	Visit E, go to F
6	F	Visit F, go to C
7	C	Visit C, traversal complete

DFS Result: A → B → D → E → F → C

3. Vertex Search Trace

Comparison	Vertex Checked	Result
1	A	Not found
2	B	Not found
3	C	Not found
4	D	Not found
5	E	Found

Result: Vertex E found after 5 comparisons.

4. Edge-Checking Trace

Representation	Check	Operations	Result
Adjacency Matrix	Matrix[E][F] = 1	1	E-F exists
Adjacency List	E → B → F	2	E-F exists

5. Execution Results Summary

Operation	Adjacency Matrix	Adjacency List
BFS neighbor checks	36	12
DFS neighbor checks	36	12
Search for E	5 comparisons	5 comparisons
Check E-F edge	1	2